

Diffusivity of H⁺ ions in saturated Kaolin clay : Fick's law-based estimations & experimental setup

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Introduction

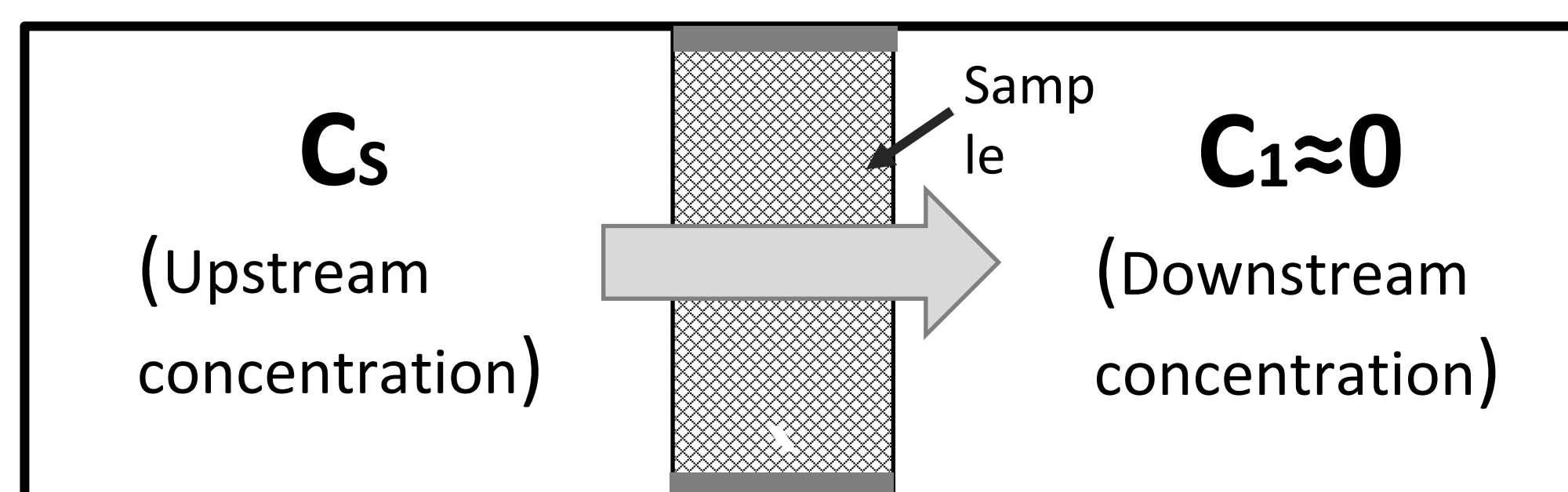
CO₂ sequestration can acidify reservoir brine, allowing H⁺ ions to diffuse into clay-rich caprock and potentially weaken its sealing capacity.

Objectives

Estimate H⁺ diffusivity in saturated Kaolin clay using through-diffusion experiments with Fick's law based numerical simulations as a guide.

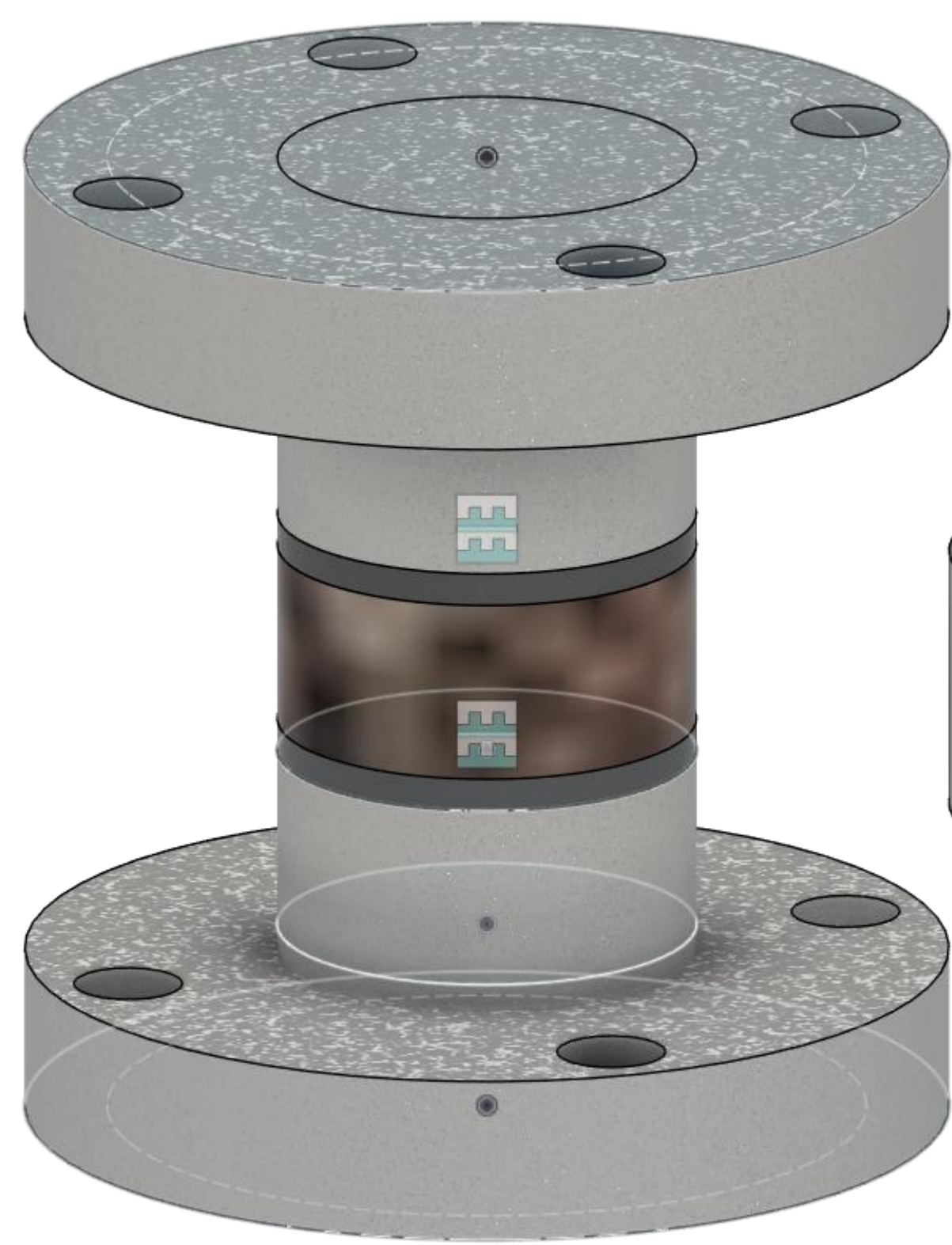
Experimental and Numerical Methods

Through diffusion apparatus set-up



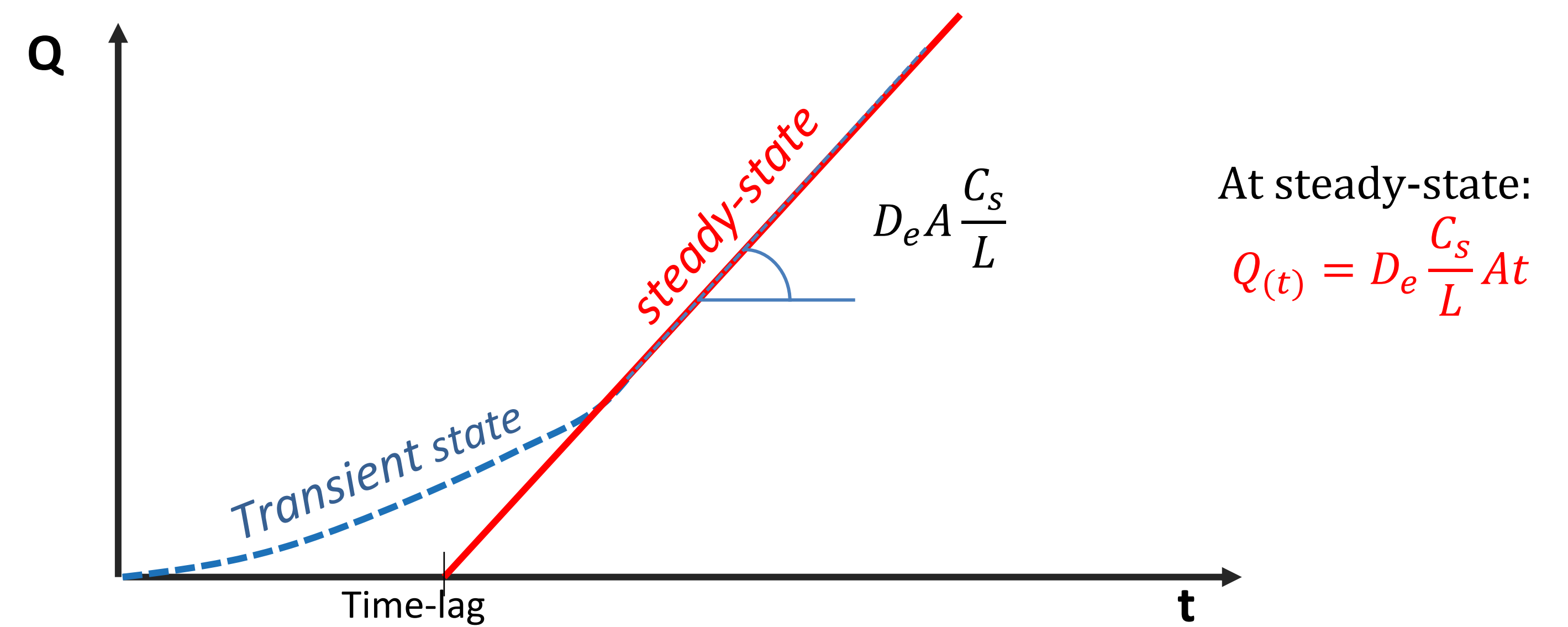
Fick's law:
 $\vec{J} = -D_e \nabla C$

At steady-state flux :
 $\nabla C = \frac{0 - C_s}{L} = \frac{-C_s}{L}$
 $J = D_e \frac{C_s}{L}$



Current status
Experiment pending;
reserved for future work

Cummulative quantity in the downstream compartment



Numerical Simulation

1 Ions balance in clay sample

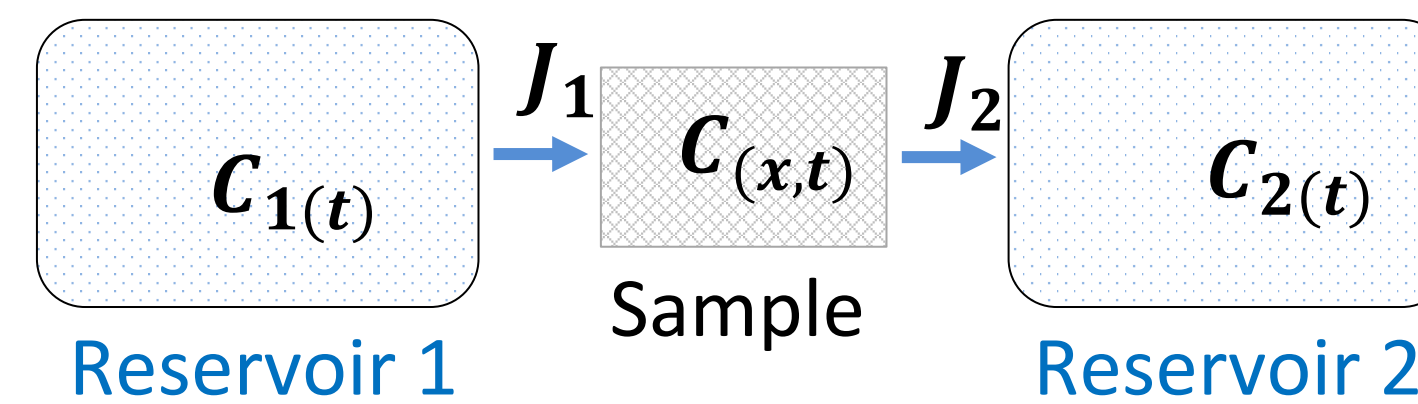
$$\frac{\partial(\theta C)}{\partial t} - \nabla \cdot \vec{J} = 0 \quad \forall x \in]0, L[$$

2 Weak Formulation

$$a_{(c,q)} = \int \frac{\phi}{\Delta t} c q \, d\Omega + \int \frac{\phi D}{\tau} \nabla c \cdot \nabla q \, d\Omega$$

$$L_{(q)} = \int \frac{\phi}{\Delta t} c_0 q \, d\Omega$$

Coupled Reservoir System



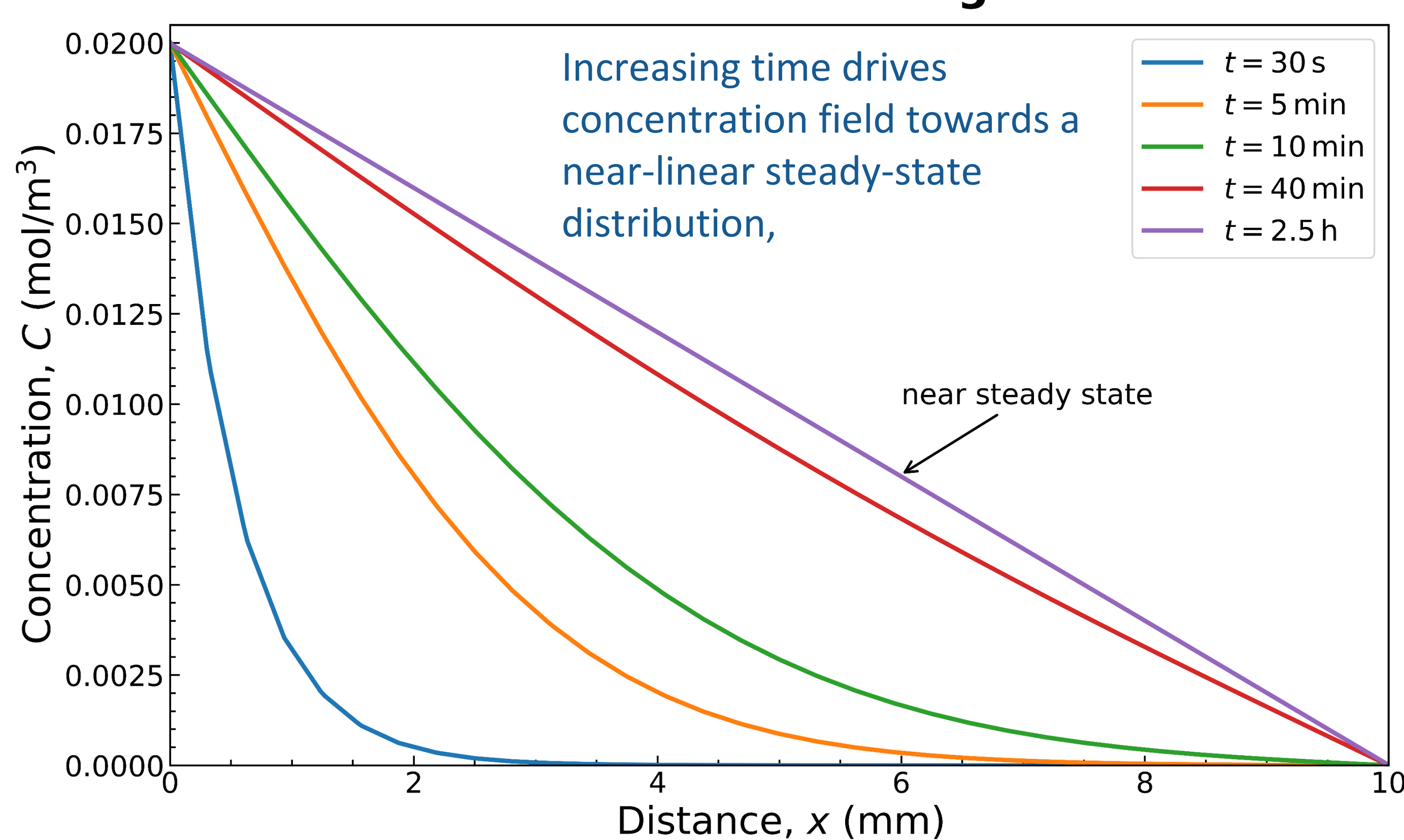
3 Reservoir Ions balance equations

$$\frac{dC_{1(t)}}{dt} V_1 = \int \vec{J}_1 \cdot \vec{n} \, ds$$

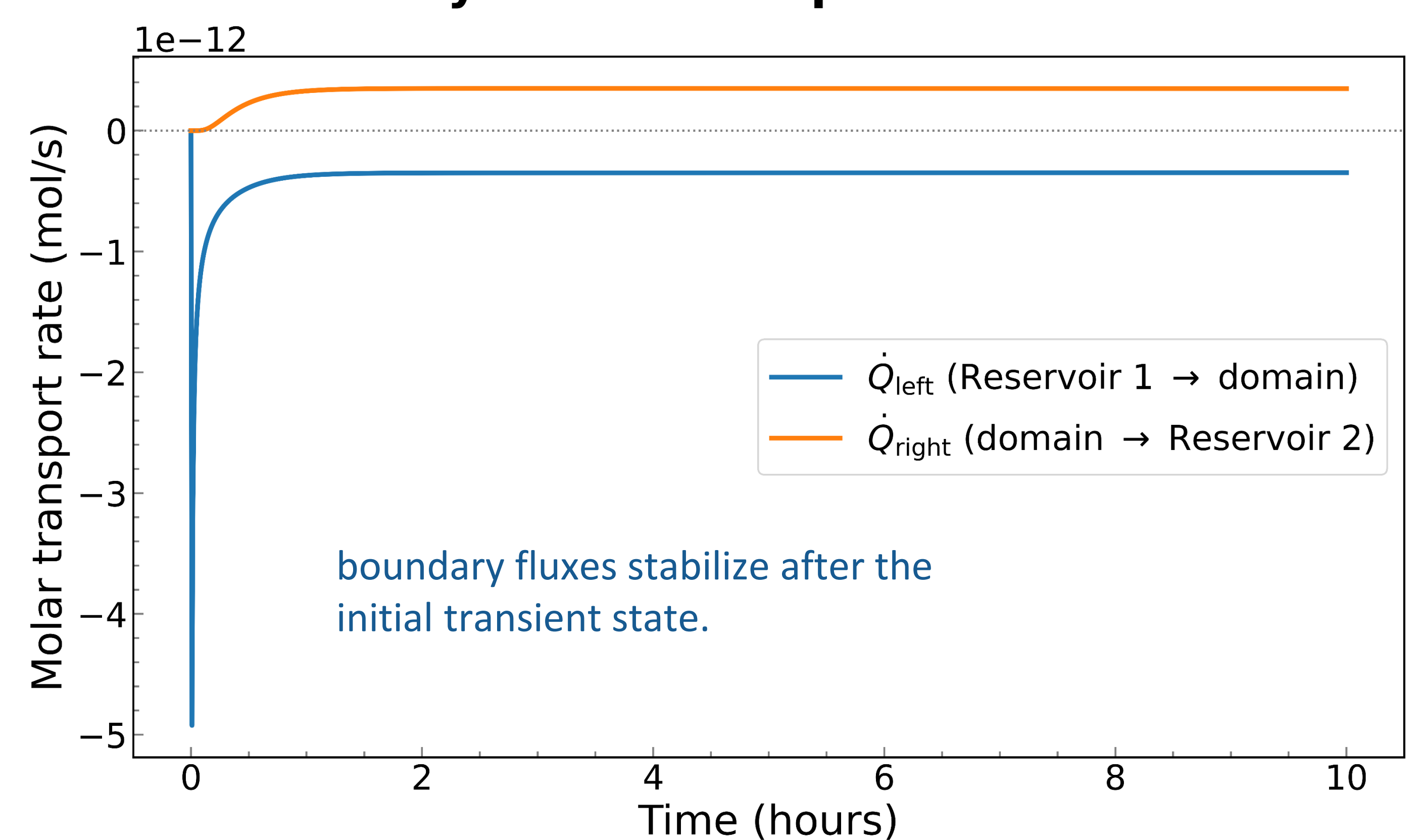
$$\frac{dC_{2(t)}}{dt} V_2 = \int \vec{J}_2 \cdot \vec{n} \, ds$$

Results

Concentration Profiles Through the Domain



Boundary Molar Transport Rates vs Time



Conclusion and Future work

- The transport process is characterized by an early transient state, after which the concentration field gradually approaches a near-steady profile while the boundary transport rates stabilize.

Current limitation and future work

- The experimental through-diffusion study could not be completed, therefore, the measurement of D_e and validation are reserved for future work.

- [1] Patel_etal_2016_Diffusivity of saturated ordinary Portland cement-based materials – A critical review of experimental and analytical modelling
[2] Tachi_Yotsuji_2014_Diffusion and sorption of Cs, Na, I and HTO sodium montmorillonite as a function of porewater salinity
[3] VanLoon_etal_2003_Diffusion of HTO 36 CL and 125 I in Opalinus Clay samples from Mont Terri Effect of confining pressure